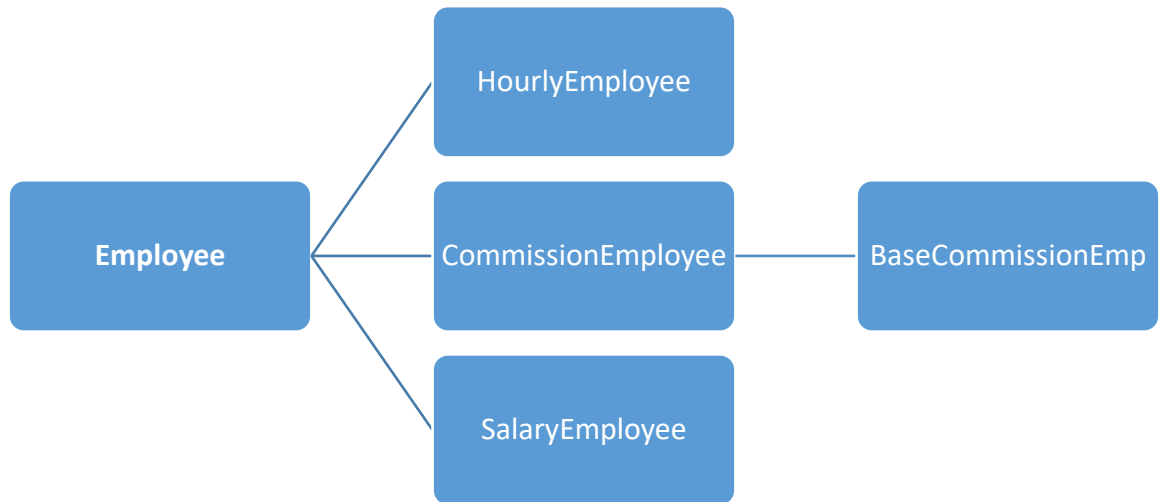


لدينا لصف موظف يورث الصف موظف باجر ساعي ويورث الموظف براتب اسبوعي ويورث الموظف الذي يتقاضى عمولة والموظف الذي يتقاضى عمولة يورث الصف الذي يتقاضى راتب زائد عمولة يرثها من الاب صف عمولة.



Employee <-

HourlyEmployee

CommissionEmployee <- BaseCommissionEmp

SalaryEmployee

الحل:

```
#pragma once
#include<iostream>
using namespace std;
class Employee
{
    string fname; string lname;
    string security;
public:
    Employee(const string&, const string&, const string&);
    void setFname(string&);
    string getFname()const;
    void setLname(string&);
    string getLname()const;
    void setSecurity(string&);
    string getSecurity()const;
    virtual double earning()const = 0;
    virtual void print()const;
    ~Employee(void);
};

#include "Employee.h"

Employee::Employee(const string& f, const string& l, const string& sec)
    :fname(f), lname(l), security(sec) { }
```

```

void Employee::setFname(string& f) { fname = f; }
string Employee::getFname() const { return fname; }

void Employee::setLname(string& l) { lname = l; }
string Employee::getLname() const { return lname; }

void Employee::setSecurity(string& sec) { security = sec; }
string Employee::getSecurity() const { return security; }

void Employee::print() const
{
    cout << getFname() << " " << getLname() << " " << getSecurity() << endl;
}
Employee::~Employee(void) { }

#pragma once
#include "Employee.h"
class HourlyEmployee : public Employee
{
    double wage;
    double hours;
public:
    HourlyEmployee(const string&, const string&, const string&, double = 0, double = 0);
    void setWage(double);
    double getWage()const;

    void setHour(double);
    double getHour()const;
    virtual double earning()const;
    virtual void print()const;
    ~HourlyEmployee(void);
};

#include "HourlyEmployee.h"
HourlyEmployee::HourlyEmployee(const string& f, const string& l,
    const string& sec, double w, double h) :Employee(f, l, sec)
{
    wage = (w < 0 ? 0 : w);
    hours = (h >= 0 && h <= 160) ? h : 0;
}
void HourlyEmployee::setWage(double w) { wage = w; }
double HourlyEmployee::getWage() const { return wage; }
void HourlyEmployee::setHour(double h) { hours = h; }
double HourlyEmployee::getHour() const { return hours; }
double HourlyEmployee::earning() const
{
    if (hours <= 40) return wage * hours;
    else return 40 * wage + (hours - 40) * wage * 1.5;
}
void HourlyEmployee::print() const{
    cout << "\nHourly Employee.. "; Employee::print();
}
HourlyEmployee::~HourlyEmployee(void) { }

#pragma once
#include "Employee.h"
class CommissionEmployee : public Employee
{
    double sales; //weekly

```

```

        double rate;    //percentage..
public:
    CommissionEmployee(const string&, const string&, const string&,
        double = 0, double = 0);
    void setRate(double);
    double getRate()const;

    void setSales(double);
    double getSales()const;
    virtual double earning()const;
    virtual void print()const;
    ~CommissionEmployee(void);

};

#include "CommissionEmployee.h"

CommissionEmployee::CommissionEmployee(const string& f, const string& l,
    const string& sec, double sal, double r) :Employee(f, l, sec)
{
    sales = sal; rate = r;
}
void CommissionEmployee::setRate(double r) { rate = r; }
double CommissionEmployee::getRate() const { return rate; }
void CommissionEmployee::setSales(double sal) { sales = sal; }
double CommissionEmployee::getSales()const { return sales; }
double CommissionEmployee::earning() const { return getRate() * getSales(); }
void CommissionEmployee::print() const
{
    cout << "\nCommission Employee.."; Employee::print();
}
CommissionEmployee::~CommissionEmployee(void) {}

#pragma once
#include "CommissionEmployee.h"
class BaseCommissionEmp : public CommissionEmployee
{
    double baseSalary;
public:
    BaseCommissionEmp(const string&, const string&, const string&,
        double = 0, double = 0, double = 0);
    void setSalary(double);
    double getSalary()const;

    virtual double earning()const;
    virtual void print()const;
    ~BaseCommissionEmp(void);
};

#include "BaseCommissionEmp.h"
BaseCommissionEmp::BaseCommissionEmp(const string& f, const string& l,
    const string& sec, double gsal, double rat, double base)
    :CommissionEmployee(f, l, sec, gsal, rat){ setSalary(base);}
void BaseCommissionEmp::setSalary(double sal) { baseSalary = sal; }

double BaseCommissionEmp::getSalary() const { return baseSalary; }
double BaseCommissionEmp::earning() const
{
    return getSalary() + CommissionEmployee::earning();
}
void BaseCommissionEmp::print() const
{

```

```

        cout << "\nBase Salary Commission Employee.. ";
    }
    BaseCommissionEmp::~BaseCommissionEmp(void) { }

```

```

#pragma once
#include "Employee.h"
class SalaryEmployee : public Employee
{
    double weekSalay;
public:
    SalaryEmployee(const string&, const string&, const string&, double = 0);
    void setweekSalary(double);
    double getweekSalary()const;

    virtual double earning()const;
    virtual void print()const;
    ~SalaryEmployee(void);
};

```

```

#include "SalaryEmployee.h"

SalaryEmployee::SalaryEmployee(const string& f, const string& l,
    const string& sec, double sal) :Employee(f, l, sec)
{
    weekSalay = sal;
}
void SalaryEmployee::setweekSalary(double sal) { weekSalay = sal; }
double SalaryEmployee::getweekSalary() const { return weekSalay; }

double SalaryEmployee::earning() const { return getweekSalary(); }

void SalaryEmployee::print() const { Employee::print(); }
SalaryEmployee::~SalaryEmployee(void) {}

```

```

#include<iostream>
#include "HourlyEmployee.h"
#include "CommissionEmployee.h"
#include "SalaryEmployee.h"
#include "BaseCommissionEmp.h"
using namespace std;
void main()
{
    HourlyEmployee hourly("Hussin", "hh", "sec", 2.3, 4.2);
    CommissionEmployee comm("Hussin", "hh", "sec", 4, 4);
    SalaryEmployee salary("Hussin", "hh", "sec", 44);
    BaseCommissionEmp baseComm("Hussin", "hh", "sec", 4, 5, 7);
    hourly.print();
    comm.print();
    salary.print();
    baseComm.print();
    cout << baseComm.earning() << endl;
}

```